

Multigrid for FEEC using Mass-Lumping and Transforming Smoothers: Algorithms and Results

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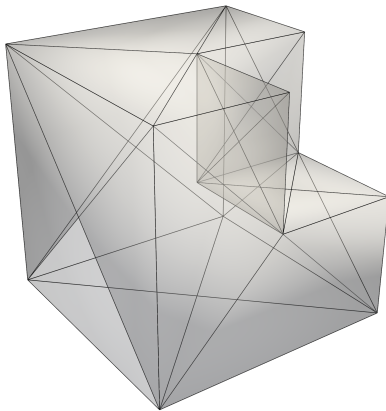
SAM, D-MATH, ETH Zürich

Toy Problem: Magnetostatics

Magnetic potential u , with
 $\operatorname{div} j = 0$

$$\begin{aligned}\operatorname{curl} \operatorname{curl} u &= j \\ -\operatorname{div} u &= 0\end{aligned}$$

Topologically trivial domain



Weak Formulation

Seek $u \in \dot{H}(\text{curl})$, $\sigma \in \dot{H}^1$ for which

$$\begin{aligned}(\text{curl } u, \text{curl } v) + (\text{grad } \sigma, v) &= (j, v) & \forall v \in \dot{H}(\text{curl}) \\(u, \text{grad } \tau) &= 0 & \forall \tau \in \dot{H}^1\end{aligned}$$

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$$(u, \text{grad } \tau) = 0 \quad \forall \tau \in \mathring{H}^1$$

Discrete de Rham Subcomplex

$$\begin{array}{ccccc} \mathring{\text{Lag}}_h & \xrightarrow{\text{grad}} & \mathring{\text{ND}}_h & \xrightarrow{\text{curl}} & \mathring{\text{RT}}_h \\ \cup & & \cup & & \cup \\ \mathring{H}^1 & & \mathring{H}(\text{curl}) & & \mathring{H}(\text{div}) \end{array}$$

Lagrange (Nodes)

Nédélec (Edges)

Raviart-Thomas (Faces)

Galerkin Formulation

Find $u_h \in \mathring{\mathbf{N}}\mathbf{D}_h \subset \mathring{H}(\text{curl})$, $\sigma_h \in \mathring{\mathbf{L}}\mathbf{a}g_h \subset \mathring{H}^1$ s.t.

$$(\text{curl } u_h, \text{curl } v_h) + (\text{grad } \sigma_h, v_h) = (j, v_h) \quad \forall v_h \in \mathring{\mathbf{N}}\mathbf{D}_h$$

$$(u_h, \text{grad } \tau_h) = 0 \quad \forall \tau_h \in \mathring{\mathbf{L}}\mathbf{a}g_h$$

Saddle-point system $\mathbb{G}_h$:

$$\mathbb{G}_h \equiv \begin{pmatrix} \mathbf{C}^T \mathbf{M}_{\text{RT}} \mathbf{C} & \mathbf{M}_{\text{ND}} \mathbf{G} \\ \mathbf{G}^T \mathbf{M}_{\text{ND}} & \mathbf{0} \end{pmatrix} \quad \text{“} \begin{pmatrix} \text{curl curl} & \text{grad} \\ -\text{div} & 0 \end{pmatrix} \text{”}$$

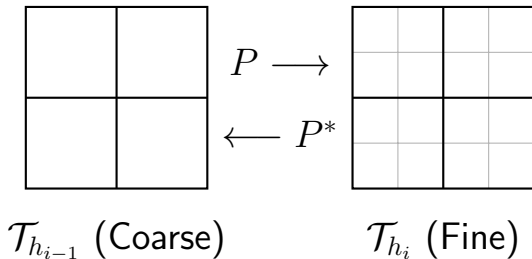
$\mathbf{G}, \mathbf{C}$: Incidence matrices representing $\text{grad}_h, \text{curl}_h$

Geometric Multigrid

Nested Hierarchy

$$\mathcal{T}_{h_0} \subset \cdots \subset \mathcal{T}_{h_L}$$

- Smoother
- Coarse Grid Correction



Transforming Smoothers: Continuous Level

Right-multiply the **System** by a **Transformation**

$$\begin{pmatrix} \text{curl curl} & \text{grad} \\ -\text{div} & 0 \end{pmatrix} \begin{pmatrix} I & \text{grad} \\ -\text{div} & 0 \end{pmatrix} = \begin{pmatrix} \text{curl curl} - \text{grad div} & 0 \\ -\text{div} & -\text{div grad} \end{pmatrix}$$

$$\text{curl curl grad} = 0 = \text{curl grad}$$

Diagonal blocks are SPD (Laplacians)

Domain Mismatch

Galerkin maps go to the dual!

$$W_h := \mathring{\mathbf{N}}\mathbf{D}_h \times \mathring{\mathbf{L}}\mathbf{a}\mathbf{g}_h$$

$$\mathbb{G}_h : W_h \longrightarrow W'_h$$

$$\mathbb{T}_h : W_h \longrightarrow W'_h$$

$\implies \mathbb{G}_h \mathbb{T}_h$ is **undefined**

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Riesz map

$$\mathbf{R}_h : W'_h \xrightarrow{\sim} W_h$$

$$\mathbf{R}_h \mathbb{G}_h \mathbf{R}_h \mathbb{T}_h : W_h \rightarrow W_h$$

Mass-Lumping

Exact Riesz map $R_h \equiv \mathbf{M}^{-1}$ dense

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$$\widetilde{\mathbf{M}} \approx \mathbf{M} \quad (\text{Diagonal})$$

$$\implies \widetilde{\mathbf{M}}^{-1} \equiv \widetilde{R}_h \text{ explicit, sparse, inconsistent}$$

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Mass-Lumped Galerkin matrix

$$\widetilde{\mathbb{G}}_h \equiv \begin{pmatrix} \mathbf{C}^T \widetilde{\mathbf{M}}_{\text{RT}} \mathbf{C} & \widetilde{\mathbf{M}}_{\text{ND}} \mathbf{G} \\ \mathbf{G}^T \widetilde{\mathbf{M}}_{\text{ND}} & \mathbf{0} \end{pmatrix}$$

Discrete Transformation

$$\begin{aligned}
 \widetilde{\text{curl}}_h^* &:= \widetilde{R}_{\text{ND}} \text{curl}'_h \widetilde{R}_{\text{RT}}^{-1} & : \mathring{\text{RT}}_h &\rightarrow \mathring{\text{ND}}_h \\
 -\widetilde{\text{div}}_h &:= \widetilde{R}_{\text{Lag}} \text{grad}'_h \widetilde{R}_{\text{ND}}^{-1} & : \mathring{\text{ND}}_h &\rightarrow \mathring{\text{Lag}}_h
 \end{aligned}$$

Discrete Transformation

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$$\widetilde{\mathbf{G}}_h := \widetilde{\mathbf{R}}_h \widetilde{\mathbf{G}}_h = \begin{pmatrix} \widetilde{\text{curl}}_h^* \text{curl}_h & \text{grad}_h \\ -\widetilde{\text{div}}_h & 0 \end{pmatrix}, \quad \widetilde{\mathbf{T}}_h := \begin{pmatrix} \mathbf{I} & \text{grad}_h \\ -\widetilde{\text{div}}_h & 0 \end{pmatrix}$$

$$\widetilde{\mathbf{G}}_h \widetilde{\mathbf{T}}_h = \begin{pmatrix} \widetilde{\text{curl}}_h^* \text{curl}_h - \text{grad}_h \widetilde{\text{div}}_h & 0 \\ -\widetilde{\text{div}}_h & -\widetilde{\text{div}}_h \text{grad}_h \end{pmatrix}$$

Distributive Gauss-Seidel Relaxation¹

$$\tilde{G}_h^{-1} = \begin{pmatrix} \mathbf{I} & \text{grad}_h \\ -\widetilde{\text{div}}_h & 0 \end{pmatrix} \begin{pmatrix} \widetilde{\text{curl}}_h^* \text{curl}_h - \text{grad}_h \widetilde{\text{div}}_h & 0 \\ -\widetilde{\text{div}}_h & -\widetilde{\text{div}}_h \text{grad}_h \end{pmatrix}^{-1}$$

¹Brandt and Dinar (1979), Hackbusch (1985, Chapter 11.3), Wittum (1990)

Distributive Gauss-Seidel Relaxation¹

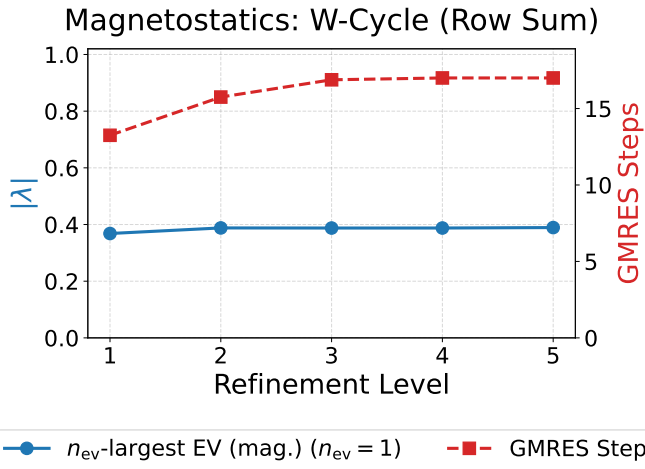
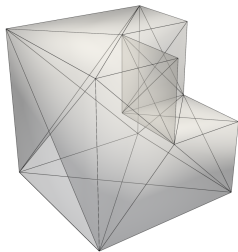
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One step of the smoother for RHS $\tilde{G}_h u = g$:

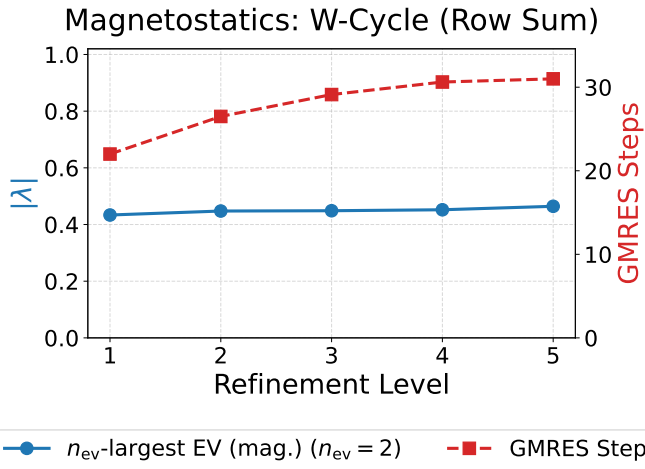
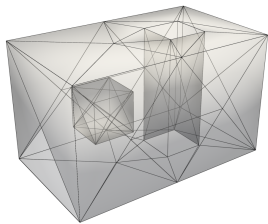
$$x \leftarrow x + \begin{pmatrix} \mathbf{I} & \text{grad}_h \\ -\widetilde{\text{div}}_h & 0 \end{pmatrix} \text{tril} \left[\begin{pmatrix} \widetilde{\text{curl}}_h^* \text{curl}_h - \text{grad}_h \widetilde{\text{div}}_h & 0 \\ -\widetilde{\text{div}}_h & -\widetilde{\text{div}}_h \text{grad}_h \end{pmatrix} \right]^{-1} (g - \tilde{G}_h u)$$

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Numerical Results: 3D Corner



Numerical Results: 3D Non-trivial Topology



Conclusion and Outlook

Further Applications

- Hodge-Dirac
- Hodge-Laplacian

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Further Topics

- Varying/Jumping Coefficients
- Rigorous Convergence Theory

Thank you for your attention!

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<https://arxiv.org/abs/2606.17934>

